

Learning goals: Basics of shell scripting continued: `exit`, `if-then-fi` construct/statement

NOTE: Please do all your work in a new folder called `lab10`. Save the script for each exercise separately as `question1.sh`, `question2.sh` and so on. At the end of the lab, please 'push' your scripts to GitHub!

EXERCISES

- (1) Write a shell script that prompts the user to enter their age, and then checks if the age is greater than or equal to 18. If it is, print "You are an adult", otherwise print "You are a minor." (Note: the logical condition to test here will use `-ge`)
- (2) Write a shell script to check if a file exists, and then to check if the file is executable. For each possibility add an `echo` statement that reflects the status.
- (3) In the above script, add `exit 200` for the if-possibility where the file exists, and `exit 201` for the possibility where the file does not exist.
 - (i) How do you check if the correct exit code is generated?
 - (ii) If you give the command `echo $?` after the if-statement, do you get the output? Why? Why not?
 - (iii) If you give the above command after `exit 200` or after `exit 201` do you get the output? Why? Why not?
- (4) Write a shell script that prompts a user to enter a number and check if its positive, negative or zero and prints a message accordingly.
- (5) Write a shell script that categorizes a user-input score (0-100) into letter grades: "A" (90+), "B" (80-89), "C" (70-79) or "Fail" (under 70).
- (6) Try the following script and fix the errors!

```
#!/bin/bash
echo "Input a number: "
read n

if [[ "$n" -gt 100 ]; then
    echo "The number is greater than 100."
else
    echo "The number is not greater than 100."
fi
if [[ "$n" -gt 100 ]] then
    echo "The number is greater than 100."
fi
```

- (7) Debug the following script:

```
#!/bin/bash
# Getting the username of the logged-in user
logged_in_user=$(whoami)
# Checking if the user is logged in
if [ -n "$logged_in_user" ];
then
    echo "The logged-in user is: $logged-in_USER"
else
    echo "User is not logged in"
fi
```